Biniyam Aschalew Tolera

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EDUCATION

KAIST, Daejeon, Republic of Korea — *Sep. 2025 – Jul. 2027 (expected)*
M.S. in Electrical Engineering — Cumulative GPA: 3.92/4.3 (96.2/100)
Advisor: Jaemin Yoo

KAIST, Daejeon, Republic of Korea — *Mar. 2021 – Aug. 2025*
B.S. in Computer Science, 🏆 *magna cum laude* — Cumulative GPA: 3.87/4.3 (95.7/100)
Minor in Industrial Engineering

RESEARCH EXPERIENCE

Data AI Lab, KAIST — *Jun. 2024 – Present*
M.S. Student (Sep. 2025 – Present); Research Intern (Jun. 2024 – Aug. 2025)
Advisor: Prof. Jaemin Yoo

- **OPAL: Structural Domain Shift in Graphs** [C1]. Developed OPAL, an unsupervised graph domain adaptation method that learns graph filters with aligned response distributions to improve transfer under structural shift.
- **Scalable In-Context Learning with Tabular Foundation Models**. Investigating tabular foundation models as in-context learners for tabular, time-series, and graph data; developing efficient context selection, with an interest in causal representations for robust in-context learning.
- **PULSE: Parameter-Efficient Graph Collaborative Filtering** [C2]. Implemented and evaluated model ablations to assess the contribution of individual components to recommendation performance.

Networking and Mobile Systems Laboratory, KAIST — *Jul. 2023 – Jun. 2024*
Research Intern | *Advisor: Prof. Sung-Ju Lee*

- **By My Eyes: Visual Prompting for Sensor Data** [C4]. Contributed to visual-prompting ideation and led the full experimental evaluation, including sensor-data pipelines, baseline implementation, and studies of prompt design and sequence-length effects.
- **SelfReplay: Adaptation of Self-Supervised Sensing Models** [C3]. Contributed to ideation for cross-domain and user adaptation; implemented baselines and designed and ran evaluations of self-supervised meta-learning.

Complexity and Real Computation Laboratory, KAIST — *Mar. 2024 – Aug. 2024*
Individual Study | *Advisor: Prof. Martin Ziegler*

- Designed experiments using the Hofstadter test to evaluate LLM agents in a virtual world.

Institute for Basic Science (IBS) — *Dec. 2022 – Feb. 2023*
Individual Study | *Advisor: Prof. Meeyoung Cha*

- Analyzed satellite imagery for Venusian climate prediction and evaluated deep learning models for weather classification.

INTERNSHIP EXPERIENCE

Bastion Robotics, Inc. — *Dec. 2023 – Feb. 2024*
AI Intern, Seoul | Developed rule-based NPC behavior and character mechanics in Unity.

BFactory Inc. — *Dec. 2022 – Feb. 2023*
Machine Learning Intern | *Seoul, Republic of Korea*

- Built an EleGANt-based makeup-transfer feature, including quality metrics, data pipelines, and fine-tuning; adapted diffusion models for personalized advertisement generation.

PUBLICATIONS

*Equal contribution

[C1] OPAL: Operator Alignment for Unsupervised Graph Domain Adaptation
Biniyam Aschalew Tolera, Akotet Yeshaw Tesema, Cheonwoo Lee, Dooho Lee, Aikta Arya, Jaemin Yoo
ACM WSDM, 2027 (accepted).

[C2] PULSE: Socially-Aware User Representation Modeling Toward Parameter-Efficient Graph Collaborative Filtering
Doyun Choi*, Cheonwoo Lee*, **Biniyam Aschalew Tolera**, Taewook Ham, Chanyoung Park, Jaemin Yoo
ACM Web Conference (WWW), 2026. **Oral presentation** [Paper] [Code]

[C3] SelfReplay: Adapting Self-Supervised Sensory Models via Adaptive Meta-Task Replay
Hyungjun Yoon, Jaehyun Kwak, **Biniyam Aschalew Tolera**, Gaole Dai, Mo Li, Taesik Gong, Kimin Lee, Sung-Ju Lee
ACM SenSys, 2025. [Paper]

[C4] By My Eyes: Grounding Multimodal Large Language Models with Sensor Data via Visual Prompting
Hyungjun Yoon, **Biniyam Aschalew Tolera**, Taesik Gong, Kimin Lee, Sung-Ju Lee
Conference on Empirical Methods in Natural Language Processing (EMNLP Main), 2024. [Paper]
Qualcomm Innovation Fellowship Korea 2025

[C5] Leveraging Synthetic Data for Data-Free Knowledge Distillation
Biniyam Aschalew Tolera, Bryan Nathanael Wijaya, Minhajur Rahman Chowdhury Mahim
Korean Software Conference (KSC), 2024. [Paper]

SELECTED PROJECTS

Synthetic Data for Zero-Shot Knowledge Distillation [C5] *Spring 2024*
KAIST CS570 course project [Code]

- Generated class-specific synthetic data by optimizing noise inputs to maximize teacher logits, enabling student training without access to the original training data.

Lang ReCAPTCHA *Summer 2023*
Junction Asia hackathon [Code] | Built a KoBERT-based CAPTCHA for low-resource language data collection and user authentication.

Fake Import Declaration Detection *Spring 2023*
3rd of 190+ participants, KAIST CS360 / Korean Customs Service competition; built model ensembles and data-processing pipelines.

HONORS & AWARDS

KAIST College of Engineering Leadership Award (Research Excellence) Mar. 2025
Selected for outstanding research by a college-wide committee from department-nominated undergraduates.

Daewoong AI and Big Data Global Scholarship Mar. 2024
KAIST Full Scholarship Mar. 2021
Hanseong-Sonjaehan Scholarship Award Mar. 2021
MindPlus National Mathematics Olympiad Gold Medalist Aug. 2019

TEACHING EXPERIENCE

Machine Learning, EAGLOPEN | *Course Instructor* Summer 2026
AI Foundation Models: Theory and Practice (EE499) | *Teaching Assistant* Spring 2026
AddisCoder Programming Summer Camp | *Teaching Assistant* Fall 2025
Introduction to Algorithms (CS300), KAIST | *Teaching Assistant* Fall 2022
Introduction to Programming (CS101), KAIST | *Teaching Assistant* Spring 2022

ACADEMIC SERVICE

Reviewer, ICML 2026 Graph Foundation Models (GFM) Workshop.
Student Volunteer, KDD 2026.

TECHNICAL SKILLS

Programming: Python, C, Java, SQL, MATLAB
Libraries and tools: PyTorch, TensorFlow, scikit-learn, NumPy, pandas, Git, Linux